A 5.5-mW +9.4-dBm IIP3 1.8-dB NF CMOS LNA Employing Multiple Gated Transistors With Capacitance Desensitization

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***Abstract*—A capacitance desensitization technique is proposed for a multiple gated transistors amplifier with source degeneration to relax second-order distortion contribution to a third-order intermodulation distortion (IMD3), as well as an induced-gate noise contribution to noise figure. An extra capacitance, which is added between gate and source nodes of input transistors in a parallel manner, can desensitize the contribution of second-order harmonic feedback to IMD3. The capacitance is useful for optimizing noise figure, as well by controlling the input matching network quality factor (Q), which can desensitize the induced-gate noise contribution to noise figure. The low-noise amplifier is implemented with the proposed technique using 1P6M 0.18-μm CMOS technology for 900-MHz code division multiple access (CDMA) receivers. It shows a third-order intercept point of +9.4 dBm and noise figure of 1.8 dB while consuming 5.5 mW at 1.5 V.**

***Index Terms*—Harmonic feedback, linearity, low-noise amplifier (LNA), multiple gated transistors (MGTRs), third-order intercept point (IIP3).**

I. Introduction

THE MAJOR parameters for RF circuits are noise figure and linearity [i.e., the third-order intercept point (IIP3)]. As CMOS scales, noise figure performance of the low-noise amplifier (LNA) improves with scaling of R_n (the equivalent noise resistance of the LNA) [1]. However, linearity does not enjoy the benefit of scaling [1]. Thus, linearity optimization continues to be one of the major issues in deep-submicrometer CMOS RF circuit design.

Several approaches have been proposed to enhance the linearity of CMOS amplifiers. The nonlinearity compensation method of transconductance is one of the most effective ways to improve linearity in practical applications [2]–[11]. Multiple gated transistors (MGTRs) uses an auxiliary transistor (AT), which is properly sized and biased to compensate for the third-order nonlinearity (g''_m, the second derivative of transconductance) of transconductance of the main transistor (MT) [4], as shown in Fig. 1(a). Fig. 1(b) shows the cancellation process of the MGTR; a negative peak of g''_m of the MT in a strong inversion region ($\mathrm{MT_bias} = V_{\mathrm{GS}}$) can be linearized with the positive peak of the AT, the bias of which is shifted by $V_{\mathrm{shift}} = 180$ mV toward a weak inversion region ($\mathrm{AT_bias} = V_{\mathrm{GS}} - V_{\mathrm{shift}}$). The threshold voltage for both transistors is 0.52 V. Thus, the composite transistor ($\mathrm{MT}+\mathrm{AT}$) has

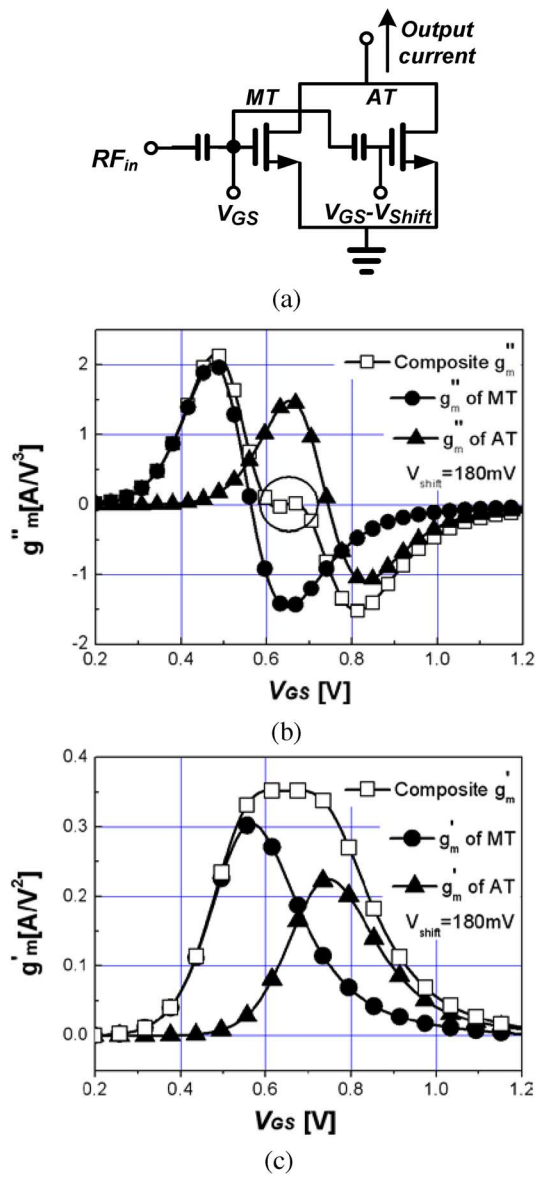

Fig. 1. (a) Schematic diagram of MGTR. (b) Illustration of g''_m cancellation using MGTR. (c) g'_m of MT, AT, and composite transistor, The composite transistor has a higher and wider g'_m than that of either the MT or AT.

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a negligible value of g_m'' in the range of 600–700 mV of $V_{\rm GS}$. However, the g_m''-linearized MGTR amplifier still suffers from a second-order distortion combined with harmonic feedback, which limits IIP3 enhancement.

Although the MGTR amplifier effectively eliminates g_m'', it has a higher and wider value of g_m' (the first derivative of transconductance) than that of the MT or AT, as shown in Fig. 1(c). Since g_m' of both the strong and weak inversion regions is positive, the composite transistor has a significant value of g_m' in the g_m'' cancelled region of $V_{\rm GS}$.

When we apply a two-tone signal at an LNA input, by virtue of the second-order distortion, second-order intermodulation distortion (IMD2) appears at the output node and is then fed back to the input through feedback paths. That feedback component and one of the input signals again experience the second-order distortion of transconductance (g_m') and generate third-order intermodulation distortion (IMD3) [2], [6]. The second-order interaction with feedback components occurs from two feedback paths in a common source (CS) topology. One is from gate-to-drain capacitance and the other is from gate-to-source capacitance. The gate-to-drain feedback is relaxed by employing a cascode topology, which improves isolation between the gate and drain node, as discussed in [2]. The gate-to-source feedback is compensated using properly chosen multiple inductors, which are implemented by an on-chip tapped inductor [6]. The multiple inductors are used to provide multiple source degeneration impedances. The multiple source degeneration inductances generate IMD3s with their own phases and magnitudes from their second-order interactions with the corresponding feedback components depending on the inductance values. The vector sum of the IMD3s from the second-order nonlinearity is then used to compensate the vector sum of the IMD3s from the third-order nonlinearity. The cascode topology applied in [2] tries to minimize the second-order contribution to IMD3 by reducing gate–drain feedback. In contrast, the multiple source degeneration method [6] tries to completely eliminate both the second- and third-order contributions to IMD3 using cancellation.

However, the tapped inductor cannot be implemented with a bond-wire inductor, which is one of the convenient ways to provide source degeneration inductance without area and noise penalties. Source degeneration using bond-wire inductor is employed in many practical applications in academia as well as industry. The elimination of the IMD3 generated by the third-order distortion contribution with another properly designed second-order distortion contribution also makes the cancellation range narrow. There exist two cancellations in that technique: one is transconductance cancellation and the other is inductor cancellation. Thus, it is hard to maintain all cancellation conditions wide enough. As a result, the wide cancellation characteristic shown in [2] no longer exists in [6].

A more convenient method is proposed to relax the harmonic feedback effect in an MGTR amplifier by desensitizing the second-order contribution to IMD3 with the addition of an extra gate–source capacitance in parallel with the intrinsic gate–source capacitance (first presented in [12]), as shown in Fig. 2. The addition of an extra gate–source capacitance was originally proposed to desensitize noise and input matching

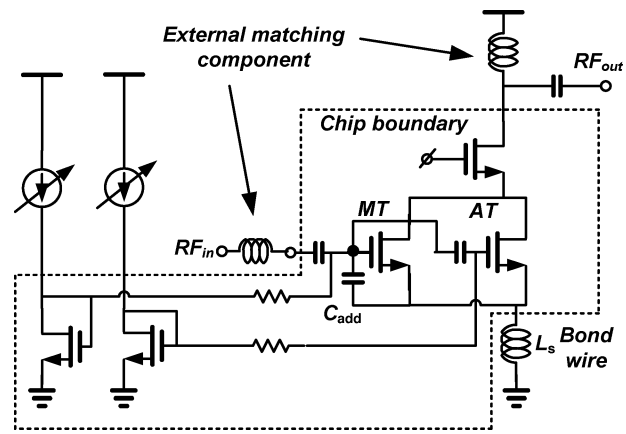


Fig. 2. Schematic of the proposed LNA.

issues from the input transistor size in a low-power LNA [13], [14]. There are other benefits to the extra gate–source capacitance, when it is combined with the MGTR. The gate–source capacitance converts second-order feedback current from source degeneration impedance to a corresponding voltage. Thus, the addition of extra gate–source capacitance can reduce the converted voltage of the second-order feedback component by reducing the gate–source impedance. Thus, it can relax the second-order distortion contribution to IMD3.

The extra gate–source capacitance along with MGTR can relax not only second-order distortion contributions to IMD3, but also induced-gate noise influence, which aggravates noise performance in a MGTR amplifier. Reference [6] shows that a transistor in a weak inversion region generates greater induced-gate noise than a transistor in a strong inversion region because the induced-gate noise is inversely proportional to the drain current in a weak inversion region. The AT in the MGTR operates in a weak inversion region so the noise figure of the MGTR is higher than the conventional one. When we employ the extra gate–source capacitance, we can relax the induced-gate noise contribution to the noise figure by bringing down the input matching quality factor (Q). Q is expressed as follows [13] when Z_1 is replaced by the inductor for input power matching:

$$Q = \frac{1}{\left(R_s + g_m \frac{L_s}{C_{\rm gs}}\right)\omega_o C_{\rm gs}} = \frac{1}{2R_s\omega_o C_{\rm gs}} \tag{1}$$

where

$$C_{\rm gs} = C_{{\rm gs}o} + C_{\rm add}. \tag{2}$$

Thus, the extra gate–source capacitance effectively relaxes the second-order distortion contribution to IMD3, as well as the induced-gate noise contribution to the noise figure. The proposed technique is especially useful for a low-power LNA design in which a small size transistor is preferred to bring down current. The induced-gate noise issue is aggravated when the input transistor is small because it inevitably increases the input matching network Q [13].

This paper is organized as follows. Following the introduction in Section I, Section II provides the linearity analysis of the extra capacitance with the MGTR. In Section III, a noise issue of the proposed method is analyzed. In Section IV, measurement results are presented. A conclusion follows in Section V.

II. Linearity Analysis

In a high-frequency region, the IMD3 of a CS amplifier is composed of two parts: one from the third-order nonlinearity of transconductance (g''_m) and the other from a second-order interaction combined with harmonic feedback. By applying the MGTR technique, we can minimize the third-order distortion contribution to IMD3, and then the second-order distortion contribution dominates IMD3 [2], [6]. Volterra series analysis provides good understanding of those behaviors.

The drain current of the CS amplifier shown in Fig. 3 can be expressed as the following Volterra series:

$$i_d(v_{\rm in}) = B_1(j\omega)\circ v_{\rm in} + B_2(j\omega_1, j\omega_2)\circ v_{\rm in}^2 + B_3(j\omega_1, j\omega_2, j\omega_3)\circ v_{\rm in}^3 + \cdots \quad (3)$$

and (3a)–(3f), shown at the bottom of this page, where B_n is the nth-order complex coefficient with magnitude and phase in Volterra expansion, and $j\omega$ is the Laplace variable. $v_{\rm in}$ is an applied input voltage. The operator "$\circ$" denotes "multiply the amplitude of each frequency component in $v_{\rm in}$ by the magnitude of the corresponding coefficient $B_n(j\omega_1, j\omega_2, j\omega_3 \ldots j\omega_n)$ and shift its phase by the phase of $B_n(j\omega_1, j\omega_2, j\omega_3 \ldots j\omega_n)$" [6], [15]–[17].

The IIP3 at $2\omega_b - \omega_a$ with the fundamental response at ω_a with a two-tone (ω_b, ω_a) excitement is given by [6], [16]

$$\text{IIP3}(2\omega_b - \omega_a) = \frac{1}{6R_s}\left|\frac{B_1(j\omega_a)}{B_3(j\omega_b, j\omega_b, -j\omega_b)}\right|. \quad (4)$$

Thus, the $B_3(j\omega_1, j\omega_2, j\omega_3)$ term, denominator of (4), explains how IMD3 is generated by the CS amplifier.

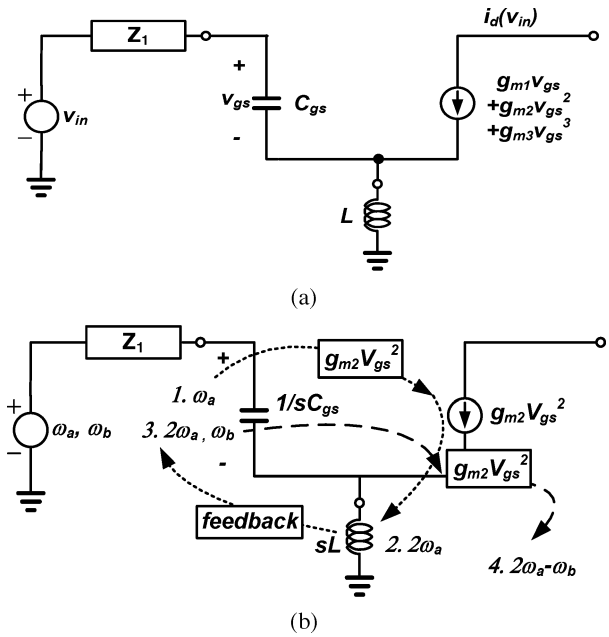


Fig. 3. (a) Equivalent circuit of proposed LNA. (b) Visual explanation of second-order interaction with harmonic feedback.

The first term of the right-hand side in (3c) relates to the third-order distortion of transconductance, as you can see g_{m3} in (3c). The $\overline{A_1(j\omega_1)A_2(j\omega_1, j\omega_2)}$ in the second term of the right-hand side in (3c) relates to a second-order interaction with harmonic feedback. In order to have more insight on the second-order contribution, we put the two-tone signal $(\omega_b, -\omega_a)$ in $\overline{A_1(j\omega_1)A_2(j\omega_1, j\omega_2)}$. Note that the two signals are close $(\omega_b - \omega_a \approx 0)$.

As shown in (3e), the second-order interaction mainly comes from 2ω (in this case, $\omega_b + \omega_b$) and $\Delta\omega(= \omega_b - \omega_a)$. The second-order nonlinearity (g_{m2}) of transconductance generates harmonic components 2ω and $\Delta\omega$ components. Those terms are then applied in $C_{\rm gs}$, which is explained by the $A_1(\Delta\omega)$ and $A_1(2\omega)$ terms in (3e) and interact with one of the input signals by the second-order nonlinearity of transconductance. However,

$$B_1(j\omega) = g_{m1}A_1(j\omega) \quad (3a)$$

$$B_2(j\omega_1, j\omega_2) = g_{m2}A_1(j\omega_1)A_1(j\omega_2) \times \{1 - g_{m1}A_1(j\omega_1 + j\omega_2)(j\omega_1 + j\omega_2)L\} \quad (3b)$$

$$\begin{aligned} B_3(j\omega_1, j\omega_2, j\omega_3,) &= g_{m3}A_1(j\omega_1)A_1(j\omega_2)A_1(j\omega_3) \times \{1 - g_{m1}A_1(j\omega_1 + j\omega_2 + j\omega_3) \times (j\omega_1 + j\omega_2 + j\omega_3)L\} \\ &\quad + \{2g_{m2}\overline{A_1(j\omega_1)A_2(j\omega_1, j\omega_2)}\} \times \{1 - g_{m1}A_1(j\omega_1 + j\omega_2 + j\omega_3)(j\omega_1 + j\omega_2 + j\omega_3)L\} \end{aligned} \quad (3c)$$

$$A_1(j\omega) = \frac{1}{1 + j\omega L g_{m1} + j\omega L \times j\omega C_{\rm gs} + Z_1(j\omega)j\omega C_{\rm gs}} \quad (3d)$$

$$\overline{A_1(j\omega_1)A_2(j\omega_1, j\omega_2)} = \frac{1}{3}\begin{bmatrix} A_1(j\omega_1)\{-g_{m2}A_1(j\omega_1)A_1(j\omega_2)A_1(j(\omega_1+\omega_2))j(\omega_1+\omega_2)L_s\} \\ +A_1(j\omega_1)\{-g_{m2}A_1(j\omega_1)A_1(j\omega_2)A_1(j(\omega_1+\omega_2))j(\omega_1+\omega_2)L_s\} \\ +A_1(j\omega_2)\{-g_{m2}A_1(j\omega_1)A_1(j\omega_1)A_1(j(\omega_1+\omega_1))j(\omega_1+\omega_2)L_s\} \end{bmatrix} \quad (3e)$$

$$g_{m1} = g_m \quad g_{m2} = \frac{1}{2}g'_m \quad g_{m3} = \frac{1}{2}g''_m \quad (3f)$$

terms that contain $\Delta\omega$ in (3e) have a trivial effect on IMD3 because $j\Delta\omega L \approx 0$. Thus, (3e) can be approximately rewritten as (5)

$$\overline{A_1(j\omega_b)A_2(j\omega_b,-j\omega_a)} = \approx \frac{1}{3}\{-g_{m2} \times |A_1(j\omega)|^2 A_1(j\omega)A_1(2j\omega)j2\omega L_s\}. \quad (5)$$

Fig. 3(b) offers a visual explanation of the above process. This process can be summarized as follows.

1) The applied two-tone signal experiences a second-order nonlinearity of transconductance and generates 2ω and $\Delta\omega$ components at the drain output current.
2) The 2ω and $\Delta\omega$ components are fed back to the gate–source capacitance by voltage–current feedback. Here, the $\Delta\omega$ component is shorted to ground with its low impedance of $j\Delta\omega L_s \approx 0$.
3) The feedback component, 2ω, and one of the applied signals, ω, again experience the second-order nonlinearity of transconductance and generate IMD3 at the drain output current.

Thus, it is important to bring down $A_1(2j\omega)$ to relax the second-order distortion contribution to IMD3 [6]. One way to reduce $A_1(2j\omega)$ is to add an extra capacitance in parallel with the gate–source capacitance. The denominator in (3d) contains a $C_{\rm gs}$ term. Thus, if we add extra capacitance to the intrinsic gate–source capacitance, as shown in Fig. 2, $A_1(2j\omega)$ can be reduced. The extra capacitor, $C_{\rm add}$, is implemented using a metal–insulator–metal (MIM) capacitor.

The IIP3 of the CS amplifier can be calculated from (4), and is given by (6), where $C_{\rm gso}$ is the intrinsic capacitance of $C_{\rm gs}$ and $C_{\rm add}$ is the extra capacitor in parallel with $C_{\rm gso}$ [6]. Equation (6) also explains the effect of third- and second-order distortion contribution to IMD3. g''_m of the denominator of (6) is negligible because of g''_m cancellation in the MGTR, thus the second term, which contains $g_m\prime$ of the denominator of (6) dominates the denominator of (6). The term containing $g_m\prime$ is divided by the $C_{\rm gs}$ term. Thus, the influence of g'_m to IIP3 is relaxed by the $C_{\rm gs}$ term. Thus, $C_{\rm add}$ can relax the second-order distortion contribution to IMD3

$$\mathrm{IIP}_3 = \frac{\frac{4}{3}g_m^2\omega^2 L_s(C_{\rm gso}+C_{\rm add})}{\left| g''_m - \frac{2g_m'^2/3}{g_m + \frac{1}{j2\omega L_s} + j2\omega(C_{\rm gso}+C_{\rm add}) + Z_1(2\omega)\frac{(C_{\rm gso}+C_{\rm add})}{L_s}} \right|}. \quad (6)$$

Fig. 4 shows a calculated IIP3 for various $C_{\rm add}$ values, as well as a simulation result. For comparative purposes, both AT-on and AT-off cases are shown. As you can see in Fig. 4, when we add $C_{\rm add}$, it will decrease the input matching network Q, and then the voltage boosting from the input matching network will also be decreased, which results in IIP3 improvement. $C_{\rm add}$ also helps the MGTR to improve IIP3. When we add $C_{\rm add}$ of 2 pF, we can obtain as large as 10-dB IIP3 improvement. As expected from (6), IIP3 increases as the $C_{\rm add}$ values increases. 2 pF is chosen for $C_{\rm add}$ values in consideration of linearity, as well as noise figure, which we discuss in Section III. The proposed technique is insensitive to a bond-wire variation because the $C_{\rm add}$ desensitizes linearity dependence on the bond-wire. Comparing L_s with $C_{\rm gso} + C_{\rm add}$, it shows that the latter has much greater admittance than the former so $C_{\rm gso}+C_{\rm add}$ dominates (6). Thus, the linearity is less sensitive to the variation of bond-wire inductance. Fig. 5 shows a simulation result of IIP3 dependence on bond-wire inductance. It shows that IIP3 is insensitive to the bond-wire variation. In our design, the bond-wire inductance is chosen to be 1 nH and the variation range is assumed to be $\pm 5\%$.

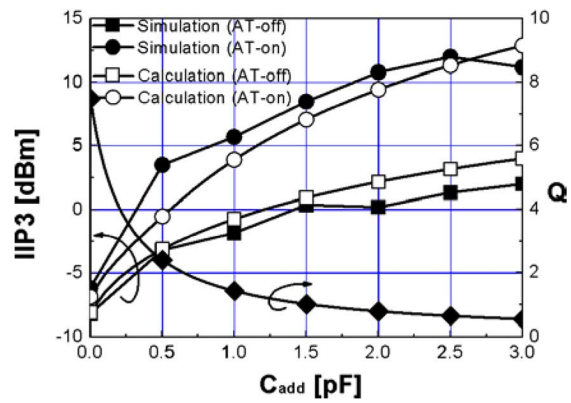


Fig. 4. IIP3 calculation and simulation result versus $C_{\rm add}$ for both AT-on and and AT-off cases.

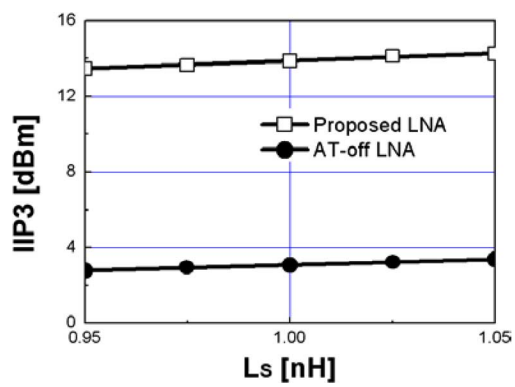


Fig. 5. IIP3 simulation result over bond-wire inductance variation ($\pm 5\%$ variation is assumed).

III. Noise Analysis

A source degeneration LNA shows excellent noise-figure performance compared to other topologies such as common gate and resistive feedback [13]. The impedance Z_1 in Fig. 3 can be replaced by the inductor for input power matching. The schematic for noise-figure analysis is given in Fig. 6. The input matching Q of Fig. 6 is shown in (1).

The high-Q input matching network can effectively reduce channel thermal noise by amplifying input signals. However, high-Q matching network can also enhance induced-gate noise [13]. Thus, the optimization of an input matching network Q is important in LNA design. The optimization is especially difficult for a low-power LNA, which requires small input transistor. Since a small transistor has small gate–source capaci-

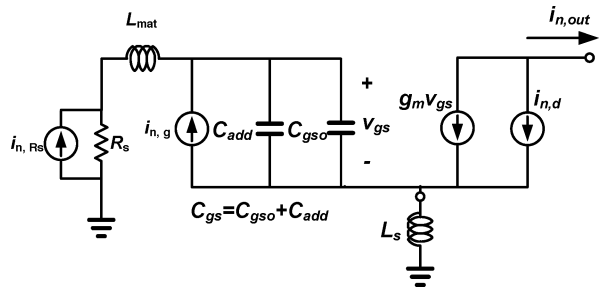


Fig. 6. Equivalent circuit of the LNA for noise-figure analysis.

tance, it then requires a high-Q input matching network. The high-Q input matching network allows the induced-gate noise to dominate the entire LNA noise. The extra capacitance can decouple the transistor size and input matching network Q [13], [14]. Thus, the extra capacitance is useful for a low-power LNA design.

A transistor in the weak inversion region shows higher induced-gate noise than a strong inversion region transistor [6]. The MGTR amplifier employs a transistor in the weak inversion region as an AT to cancel the third-order nonlinearity of the main amplifier, which is biased in the strong inversion region. Thus, the input matching network Q should be lower than that of conventional LNA design. The extra capacitance introduced in Section II has another benefit to noise-figure optimization. It can relax the effect of enhanced induced-gate noise of the transistor in the weak inversion region by lowering the Q of the input matching network.

The thermal noise of the drain current ($\overline{i_{\rm nd}^2}$) and an induced-gate noise ($\overline{i_{\rm ng}^2}$) are expressed with the following equations:

$$\overline{i_{\rm nd}^2} = 4kT\Delta f\gamma g_{d0} \tag{7}$$

$$\overline{i_{\rm ng}^2} = \frac{4kT\Delta f\beta\cdot 4(\omega C_{\rm ox}WL_{\rm eff})^2}{9g_{d0}} \tag{8}$$

where k is the Boltzmann's constant, T is the absolute temperature, γ and β are the bias-dependent noise coefficients, g_{d0} is the drain–source conductance at $V_{\rm DS}=0$, $C_{\rm ox}$ is the gate–oxide capacitance per unit area, W is the channel width, and $L_{\rm eff}$ is the channel length. The induced-gate noise in a weak inversion region is expressed with the following equation:

$$\overline{i_{\rm ng}^2} = \frac{4kT\Delta f\beta\cdot 4(\omega C_{\rm ox}WL_{\rm eff})^2}{9}\cdot\frac{\phi_t}{I_D}. \tag{9}$$

I_D is the drain saturation current and ϕ_t is the thermal voltage kT/q. Equation (9) shows that the induced-gate noise is inversely proportional to drain current and it aggravates noise-figure performance in the MGTR LNA.

The uncorrelated portion of induced-gate noise from the drain current noise of the MT is given by [6]

$$\overline{i_{{\rm ng}u}^2} = \overline{i_{\rm ng,AT}^2} + \overline{i_{\rm ng,MT}^2}(1-|c_{\rm MT}|^2) \tag{10}$$

where

$$|c_{\rm MT}| = \frac{\overline{i_{\rm ng,MT}\cdot i_{\rm nd,MT}^*}}{\sqrt{\overline{i_{\rm ng,MT}^2}\cdot\overline{i_{\rm ng,MT}^2}}}. \tag{11}$$

Since the MT is biased in strong inversion, $|c_{\rm MT}|\approx 0.4$, the correlated portion can be ignored for simplicity for a noise-figure calculation [13]. Also, we can ignore the drain current noise of the AT and the induced-gate noise of MT because $I_{\rm MT}\gg I_{\rm AT}$ [13].

The noise figure (F) is obtained at resonance

$$F = 1+\frac{\overline{i_{n,o,d}^2}+\overline{i_{n,o,g}^2}}{\overline{i_{n,o,Rs}^2}} \tag{12}$$

where

$$i_{n,o,d} = \frac{1}{2}i_{n,d} \tag{13}$$

$$i_{n,o,g} = 2g_{m,\rm TOT}R_sQ^2\left(1-j\frac{1}{2Q}\right)i_{n,g} \tag{14}$$

and (12) can be re-expressed as follows:

$$F = 1+\frac{\gamma' g_{d0,\rm MT}}{4g_{m,\rm TOT}^2R_sQ^2}+\frac{\beta'(1+4Q^2)R_s(\omega C_{\rm gso})^2}{g_{d0,\rm AT}} \tag{15}$$

where

$$\begin{aligned}\beta' = {} & \frac{g_{d0,\rm AT}C_{\rm ox}^4W_{\rm AT}^4\mu_{\rm eff}^2}{C_{\rm gso}^2I_{D,\rm AT}^3}\\ & \times\Bigg[V_{\rm GT}^2V_{\rm as}^2V_{\rm DS}-V_{\rm GT}V_{\rm as}(V_{\rm GT}+V_{\rm as})V_{\rm DS}^2\\ & +\frac{V_{\rm GT}^2+4V_{\rm GT}V_{\rm as}+V_{\rm as}^2}{3}V_{\rm DS}^3\\ & -\frac{V_{\rm GT}+V_{\rm as}}{2}V_{\rm DS}^4+\frac{V_{\rm DS}^5}{5}\Bigg]\end{aligned} \tag{15a}$$

$$\gamma' = \frac{4V_{\rm GT}^2+V_d^2+V_dV_{\rm GT}}{3V_{\rm GT}^2(V_{\rm GT}-V_d)g_{d0,\rm MT}}\alpha I_D \tag{15b}$$

$$V_{\rm as} = V_{\rm DS}-\frac{\frac{V_{\rm GT}V_{\rm DS}}{2}-\frac{V_{\rm DS}^2}{6}}{V_{\rm GT}-V_{\frac{\rm DS}{2}}} \tag{15c}$$

$$C_{{\rm gs}o} = \frac{2}{3}C_{\rm ox}WL_{\rm eff}. \tag{15d}$$

Here, $g_{m,\rm TOT}=g_{m,\rm MT}+g_{m,\rm AT}\approx g_{m,\rm MT}$ due to $g_{m,\rm MT}\gg g_{m,\rm AT}$, $V_d=I_D/(WC_{\rm ox}v_{\rm sat})$, $V_{\rm GT}=V_{\rm GS}-V_{\rm th}$, α is the bulk charge factor and $v_{\rm sat}$ is saturation velocity. γ' and β' are the bias-dependent noise coefficients for a short-channel field-effect transistor (FET). The long-channel values for γ in (7) and β in (8) are 2/3 and 4/15, respectively [13], [18]. In short-channel cases, however, the values of γ' and β' should be adjusted for a possible increase of our assumption due to short-channel effects.

TABLE I
PARAMETERS AND VALUES IN NOISE-FIGURE CALCULATION

Parameter	Value
γ'	2
β'	1.4
$g_{m,MT}$	40 mA/V
$g_{m,AT}$	1.9 mA/V
$g_{d0,MT}$	$\approx g_{m,MT}$
$g_{d0,AT}$	3.5 mA/V
W ($W_{MT} + W_{AT}$)	285 um (185+100 um)
L_{eff}	180 nm
C_{ox}	6.9 mF/m^2
μ_{eff}	0.045 m^2/V
α	1

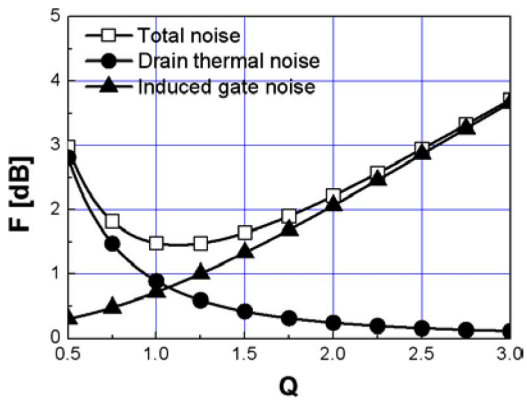


Fig. 7. Noise-figure calculation versus input matching network Q.

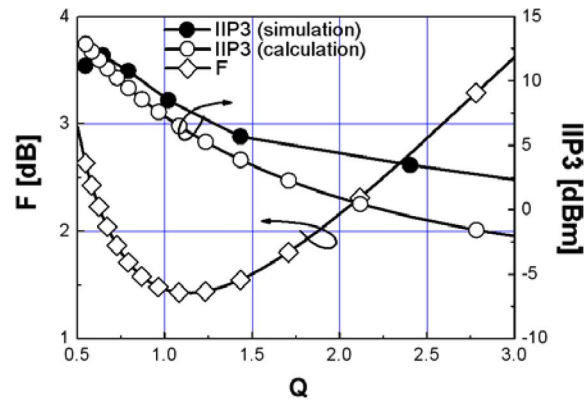


Fig. 8. Noise figure and IIP3 versus input matching network Q.

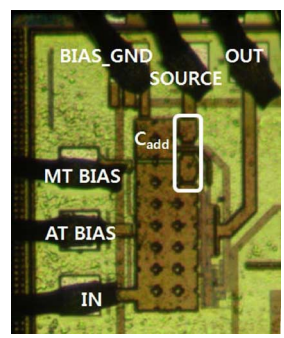


Fig. 9. Microphotograph of the proposed LNA.

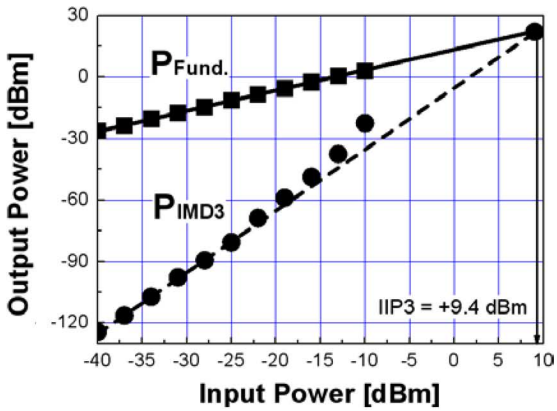
The values of γ' and β' are calculated from (15a)–(15c) and are given in Table I [19], [20].

In the noise-figure calculation, noise contribution from output resistance is neglected for simplicity. As expected, the higher input matching Q we use, the higher induced-gate noise is introduced, while higher Q decreases drain thermal noise with higher gain, as you can see in (15). Thus, the extra capacitance increases drain thermal noise contribution to noise figure and reduces gain by lowering Q, but it decreases induced-gated noise contribution to the noise figure. In the MGTR amplifier, input matching network should have a lower Q than that of a conventional amplifier because the induced-gate noise contribution is greater than that of the conventional amplifier. Fig. 7 shows the noise-figure calculation result. It shows the noise figure from the contribution of induced-gate noise only and noise figure from that of thermal noise only, as well as total noise figure.

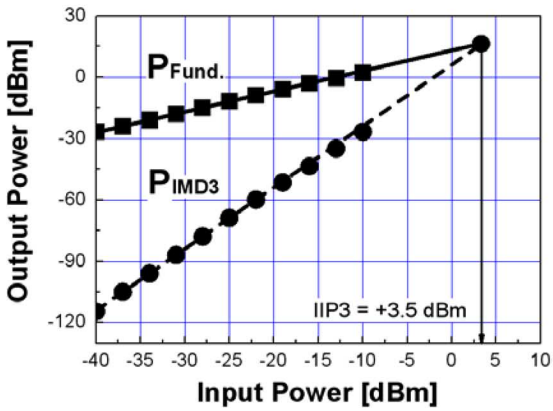
The BSIM3 model do not provide an induced-gate noise model. Thus, we try to optimize the noise figure using (15). The values of coefficients in the noise-figure calculation are shown in Table I. Even though the values are not obtained from direct measurement of 0.18-μm CMOS technology, it is enough to give an optimization guideline for noise-figure design. Fig. 7 shows the noise-figure calculation result over the input matching network Q. The noise figure, which only considers induced-gate noise, increases as Q increases, while the noise figure that only considers drain thermal noise decreases as Q increases, thus there exists optimum Q, as shown in the total noise figure. It shows that $Q \approx 1$ has the lowest noise figure. In a conventional LNA design, the input matching network Q is selected around three for low-noise application [13]. In this design, we choose $Q \approx 1$ in order to desensitize the influence of induced-gate noise on the total noise performance. It is plausible to select $Q \approx 1$ in the MGTR amplifier, which has higher induced-gate noise contribution than that of a conventional LNA. In our design, 2 pF of $C_{\rm add}$ brings the Q down to around 1 as shown in Fig. 4. Fig. 8 shows the noise figure and IIP3 versus Q. It shows $Q \approx 1$ is a good choice for both the IIP3 and noise figure.

IV. DESIGN AND MEASUREMENT RESULTS

The LNA is implemented with a 1P6M 0.18-μm CMOS process. Fig. 9 shows a microphotograph of the LNA, which occupies a silicon area of 220×330 μm^2 excluding pads. It occupies a small silicon area because the source degeneration inductor is implemented with wire-bonding. External components are used for input and output matching, as shown in Fig. 2. A current-mirror-based bias circuit is employed to reduce the influence of process, voltage, and temperature (PVT) variation to IIP3 [2]. The MT operates in a strong inversion region, and the AT operates in a weak inversion region. The widths are 185 and 100 μm, respectively. For comparative purposes, the AT can be turned off by applying zero voltage at the gate of the AT. Fig. 10(a) and (b) shows the measured IIP3 at maximum IMD3 reduction and that of AT-off LNA, respectively. IIP3 is measured by a two-tone test whose input frequencies are 900 and 901 MHz. The proposed LNA shows an IIP3 of +9.4 dBm while the AT-off LNA shows an IIP3 of +3.5 dBm. Fig. 11 shows IIP3 measurement versus AT bias, while MT bias is



(a)



(b)

Fig. 10. IIP3 measurement result of the: (a) proposed LNA and (b) AT-off LNA.

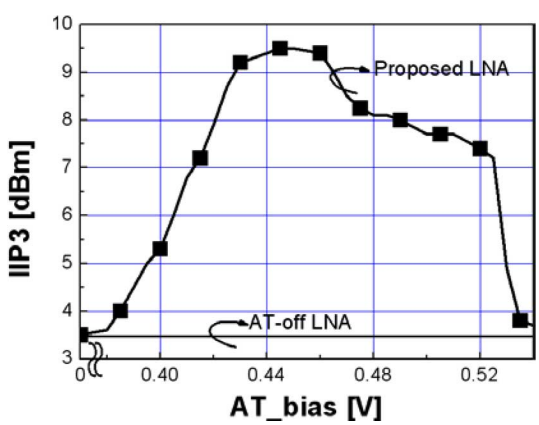


Fig. 11. IIP3 Measurement of the proposed amplifier versus AT bias.

fixed at 0.61 V. It shows maximum 6-dB IIP3 improvement over the AT bias. Fig. 12 shows IIP3 and gain over frequency at maximum IMD3 reduction.

The discrepancy between the maximum IIP3 obtained in the simulation result in Fig. 4 and that of the measurement in Fig. 10(a) possibly originates from imperfect g_m'' modeling, as discussed in [21]. Fig. 13 shows noise-figure measurement

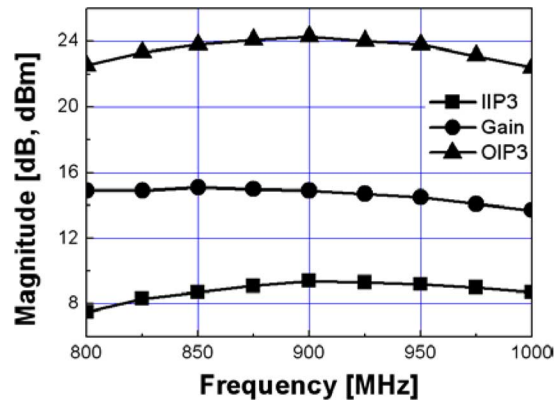


Fig. 12. IIP3 and gain measurement result versus frequency.

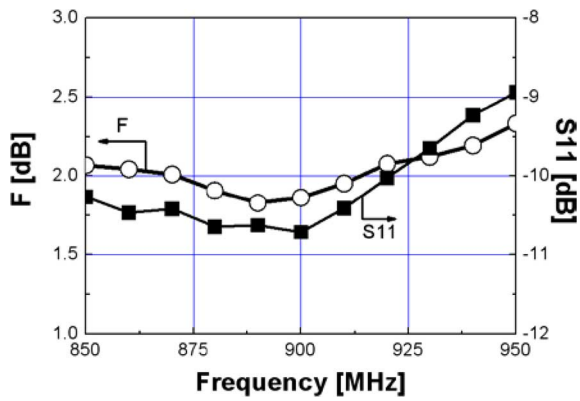


Fig. 13. $S11$ and noise-figure measurement result of the proposed amplifier.

results, as well as input matching. The noise figure is 1.8 dB at 900 MHz and input matching is below -10 dB. The AT-off LNA shows similar noise figure (1.85 dB), which indicates that induced-gate noise contribution from the AT is quite relaxed. The reason why the AT-off LNA has a slightly worse noise figure is that the gain of the AT-off LNA is smaller than that of the AT-on LNA so noise from output resistance, which was neglected in our calculation, affects noise figure. Thus, even though the AT generates higher induced-gate noise, the AT-off LNA has a slightly worse noise figure. This also shows that $C_{\rm add}$ effectively relaxes the contribution of induced-gate noise from the AT.

There exists a slight discrepancy between our calculation and measurement result since the parameters used in the calculation such as γ' and β' are not exact ones. Those values are not obtained from measurement, also layout parasitics such as interconnection loss and external components loss are not considered in the calculation.

The performance summary of the proposed LNA and comparisons with other low-power LNAs are shown in Table II. Among the many LNAs, those having below about 20-mW power consumption are selected for comparison, and our result shows the highest figure of merit (F.O.M) among many low-power LNAs. F.O.M is defined by (16) where output third-order intercept

TABLE II
MEASUREMENT SUMMARY AND COMPARISON OF PROPOSED LNA WITH OTHER LOW-POWER LNAs

Work	Tech.	Freq [GHz]	IIP3 [dBm]	F [dB]	Gain [dB]	P_{dc} [mW]	F.O.M
This work	**0.18μm CMOS**	**0.9**	**+9.4**	**1.8**	**14.9**	**5.55**	**94**
AT-off LNA	0.18μm CMOS	0.9	+3.5	1.85	14.3	5.33	21
'08 MWCL [22]	0.13μm CMOS	2.14	-5.3	1.97	16.6	7.5	3
'00 ESSCIRC [23]	0.35μm CMOS	0.9	0	0.9	14	18	6
'06 MWCL [24]	0.18μm CMOS	2	13.3	1.4	12.8	14.4	74
'01 JSSC [25]	0.35μm CMOS	0.9	1	0.85	15	17.6	10
'04 TMTT [26]	0.25μm CMOS	0.9	-4	1.35	12	2	9
'00 JSSC [27]	0.25μm CMOS	0.9	-15	3.3	25	0.75	12
'02 JSSC [28]	0.24μm CMOS	0.8	7.1	0.9	8.8	7.5	23
'09 ISSCC [29]	0.18μm CMOS	0.3 -0.92	-3.2	2	21	3.6	29
'08 JSSC [30]	0.13μm CMOS	5	-10	2.1	20	3.8	4
'08 JSSC [31]	0.13μm CMOS	0.8 -2.1	16	2.6	14.5	14.5	79

point (OIP3) is the output-referred IIP3, F is the noise figure, and $P_{\rm dc}$ is the power consumption

$$\text{F.O.M} = \frac{\text{OIP3(mW)}}{(F-1)P_{\rm dc}(\text{mW})}. \tag{16}$$

CDMA requirements for single-tone desensitization demand greater higher than +8 dBm of IIP3 for LNA [6], [7]. Our result of +9.4 dBm is enough for CDMA application. Our design target is to minimize the power consumption and noise figure, while maintaining the required IIP3. As shown in Table II, our result yields the lowest power consumption with a good noise figure while achieving the CDMA IIP3 requirement.

V. CONCLUSION

The capacitance desensitization technique is useful to relax a second-order distortion contribution to IMD3, as well as an induced-gate noise to noise figure in the source-degenerated MGTR amplifier. Volterra series analysis shows that extra capacitance in parallel with the intrinsic gate–source capacitance can reduce the second-order amplification of the second-order harmonic feedback component (2ω). Thus, it can maintain high linearity. Also, increased induced-gate noise in the MGTR amplifier because of AT in the weak inversion region is properly controlled by controlling the input matching network Q with the extra capacitance. The technique is especially useful for low-power high-linearity LNA design, which needs a small size input transistor. Measurement results shows an IIP3 of +9.4 dBm, noise figure of 1.8 dB, and gain of 14.9 dB while consuming 5.5 mW at 1.5-V supply voltage.

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